daughters: would she on average contribute more to the gene pool of the future than a rival mother who invested in daughters? Do genes for preferring sons become more or less numerous than genes for preferring daughters? What Fisher showed is that under normal circumstances the stable sex ratio is 50:50. In order to see why, we must first know a little bit about the mechanics of sex determination.

In mammals, sex is determined genetically as follows. All eggs are capable of developing into either a male or a female. It is the sperms that carry the sex-determining chromosomes. Half the sperms produced by a man are female-producing, or X-sperms, and half are male-producing, or Y-sperms. The two sorts of sperms look alike. They differ with respect to one chromosome only. A gene for making a father have nothing but daughters could achieve its object by making him manufacture nothing but X-sperms. A gene for making a mother have nothing but daughters could work by making her secrete a selective spermicide, or by making her abort male embryos. What we seek is something equivalent to an evolutionarily stable strategy (ESS), although here, even more than in the chapter on aggression, strategy is just a figure of speech. An individual cannot literally choose the sex of his children. But genes for tending to have children of one sex or the other are possible. If we suppose that such genes, favouring unequal sex ratios, exist, are any of them likely to become more numerous in the gene pool than their rival alleles, which favour an equal sex ratio?

Suppose that in the elephant seals mentioned above, a mutant gene arose that tended to make parents have mostly daughters. Since there is no shortage of males in the population, the daughters would have no trouble finding mates, and the daughter-manufacturing gene could spread. The sex ratio in the population might then start to shift towards a surplus of females. From the point of view of the good of the species, this would be all right, because just a few males are quite capable of providing all the sperms needed for even a huge surplus of females, as we have seen. Superficially, therefore, we might expect the daughter-producing gene to go on spreading until the sex ratio was so unbalanced that the few remaining males, working flat out, could just manage. But now, think what an enormous genetic advantage is enjoyed by those few parents who have sons. Anyone who invests in a son has a very good chance of being the grandparent of hundreds of seals. Those who are producing nothing but daughters are assured of a safe few grandchildren,